A Discussion of Phase Cells as the Basic Descriptive Units of Structure in Statistical Mechanics

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Februrary 03, 2026

Abstract

From the perspective of structure ontology, phase cells are taken as the basic units for describing structure in statistical mechanics. Within this framework, the relationship between a system's temperature and the distribution of energy over phase cells is one of representation and being represented; that is, temperature is a statistical representation of the system's energy distribution. Consequently, the photon number in a thermal-equilibrium radiation system has statistical significance and whether it is conserved is not a question that needs to be judged; rather, the process of temperature increase in the system, in a statistical sense, represents the redistribution of energy over phase cells.

Introduction

When describing a physical phenomenon, one usually first chooses a model, for example, regarding a system as being composed of particles or of waves. Different choices of models lead to different theoretical starting points, and thus the interpretations of the same phenomenon may differ.

Statistical mechanics provides a way of thinking about research problems in which neither physical particles nor waves are taken as the primary starting point of description; instead, structure is taken as the primary object of consideration. For convenience of expression, we refer to theories that take physical particles as the starting point of description as *particle ontology*, and to theories that take structure as the starting point of description as *structure ontology*. From the perspective of structure ontology, phase cells are taken as the basic descriptive units.

Temperature, in thermodynamics, is usually regarded as a macroscopic parameter, and in most cases, treated as an a priori quantity. With the establishment of the relation $\frac{1}{T} = \frac{\partial S}{\partial E}$ in the fundamental equations of thermodynamics, temperature is expressed through the mathematical relationship between entropy and internal energy, and here temperature is no longer an a priori quantity. Thereafter, with the development of statistical mechanics, temperature appears in statistical description of energy. Accordingly, a basic question naturally arises: what position does temperature occupy in statistical physics?

From the perspective of structure ontology, temperature represents the distribution of the system's energy over phase cells, and the temperature has statistical significance. On this basis, two phenomena can be described: the phenomenon that the total number of

photons in a thermal-equilibrium radiation system is not conserved, and the phenomenon of the heating process in a system. The descriptive language for these two phenomena is fundamentally different from that of particle ontology.

1 On the Choice of the Object of Study

In the study of physical problems, the construction of a theory usually begins with the specification of an object of study, followed by a description of physical phenomena. This gives rise to a physical viewpoint centered on what is taken as the object of study. From this perspective, one of the reasons for the differences between physical theories lies in the choice of their respective objects of study.

In statistical physics, the present work classifies such points of departure into two types: particle ontology and structure ontology. Particle ontology takes the Boltzmann distribution as its core and treats material particles as the objects of study, focusing on their concrete mechanical quantities, such as position and momentum. Statistical descriptions over this framework consist in counting and characterizing the microscopic states accessible to a large number of material particles. Structure ontology, by contrast, is represented by Bose distribution and establishes phase cells as the primary objects of study. Its central concern is the occupation of the total energy among different phase cells in a system. Structure ontology does not concern itself with what particles are, nor with their specific behaviors; rather, it describes the state of a system through the manner in which structures are occupied.

Starting from the perspective of structure ontology, this paper takes phase cells as the basic descriptive units of structure in statistical mechanics, and characterizes the physical state of a system by the distribution of energy among phase cells. In this sense, temperature acquires a well-defined statistical meaning.

2 The Meaning of Temperature in Statistical Mechanics

On the basis of the theoretical framework and statistical derivation presented in the previous paper [1], it has been established that phase cells are taken here as the primary objects of study. The core issue is to examine the distribution of the system's energy over phase cells. In other words, by considering how the structure—composed of phase cells—is occupied by energy, one can describe the state of the system. Here, the phase cell is the basic unit of the structure. This section will show that the physical state of a system can be characterized by the distribution of energy over phase cells; in this sense, temperature has a definite statistical meaning.

Temperature was initially associated with the concept of heat, but with the development of thermodynamics the two gradually become separated. In thermodynamics, temperature is regarded as a parameter describing the macroscopic state of a system. What, then, is the position of temperature in statistical mechanics?

From the perspective of statistical mechanics, under the constraint of total energy alone, that is, by introducing only one Lagrange multiplier. And the distribution of energy among phase cells follows the principle of maximum entropy (i.e. the principle of maximizing the number of microscopic states), yielding the average energy $\overline{E^s}$ in a phase

cell as [1, 2, 3]

$$\overline{E^s} = \frac{E^s}{e^{E^s/k_{\mathrm{B}}T} - 1}. \tag{1}$$

where the superscript s denotes the s-th phase cell, E^s is the energy carried by an quantum within the phase cell, k_{B} is the Boltzmann constant, and T denotes the temperature of the system.

Now, dividing Eq.(1) by E^s, a dimensionless result can be obtained. We denote this result by $\overline{n^s}$, it denotes the average occupation number of quanta within the s-th phase cell, hereafter referred to simply as the occupation number. Its explicit expression is

$$\overline{n^s} = \frac{1}{e^{E^s/k_{\mathrm{B}}T} - 1}. \tag{2}$$

It can be seen that the occupation number $\overline{n^s}$ is a statistical quantity. Moreover, when the occupation number changes, the temperature of the system changes as well. This may be understood as follows: the distribution of energy over phase cells and temperature are not in a causal relation, but rather in a relation of the represented and the representing. That is, temperature is a statistical representation of the distribution of the system's energy over phase cells. Put differently, the microscopic distribution of energy over the structure of the system is expressed macroscopically through the parameter of temperature.

In addition, the thermodynamic concept of temperature is determined through the partial derivative of entropy with respect to internal energy, $\frac{1}{T} = \frac{\partial S}{\partial \overline{E}}$, which, in mathematical terms, establishes the relation among temperature, entropy, and internal energy. Statistical mechanics further enriches the physical content of this relation: temperature, as a macroscopic parameter, represents the distribution of the system's internal energy over phase cells.

In the traditional thermodynamic formulation, one may think the rise of the system's temperature is directly related to the energy transferred from the external environment. In statistical mechanics, however, this question needs to be reconsidered. The energy transferred from the outside does not directly affect the temperature of the system; rather, it changes the structure of the distribution of the system's energy over phase cells. This result is then characterized through the macroscopic parameter of temperature. More specifically, after external energy is transferred to the system, the system redistributes this energy among its phase cells (the distribution follows the principle of maximum entropy), temperature is the macroscopic manifestation of the average energy in each phase cell.

3 The Statistical Meaning of Photon Number in a Thermal-Equilibrium Radiation System

In statistical physics, a thermal-equilibrium radiation system may be regarded as a photon energy field. From the perspective of the occupation of the phase cell structure by energy, it can be seen that, in a thermal-equilibrium radiation system, whether the total photon number of the system remains constant is not a question that needs to be judged. This is because photon number is a statistical quantity, and statistical quantities do not involve the question of conservation. This point will be explained in detail below.

First, as mentioned above, the occupation number is a statistical quantity. It is also a dimensionless numerical value, which only reflects the microscopic state of the system.

Second, in physics, A single photon carries an energy E (for a photon, $E = h\nu$, where ν is the photon frequency). Combining these two points, one may conclude that the occupation number and photon number are equivalent in a statistical sense. Since the occupation number is a statistical quantity, photon number may likewise be understood as a statistical quantity.

In this way, when the concept of the photon enters statistical mechanics, it loses its original meaning as a "particle". Moreover, when photon number exists as a statistical quantity used to characterize the state of a system, whether the total photon number is conserved is therefore not a question that needs to be judged here. Nor should one ask where the photons have gone when the system's temperature changes; one can only ask how photon energy is redistributed among phase cells when system's temperature changes.

4 The Statistical Understanding of the Heating Process

On the basis of the above statistical positioning of temperature, the same physical phenomenon can be understood differently. The heating of a system is taken as an example below.

From the perspective of dynamics, since the system is acted upon by the external environment, the particles undergo transitions from lower energy levels to higher energy levels. During this process, the average energy of the particles increases, and macroscopically the system appears as an increase in temperature.

However, from the perspective of statistical mechanics, the focus is no longer on the transition behavior of individual particles, but on the distribution of the system's energy among phase cells. In this statistical sense, the reason for the rise in the system's temperature is the increase in the average energy within each phase cell; the macroscopic parameter of temperature represents this change.

Therefore, in statistical mechanics, the heating process more essentially reflects the redistribution of energy over phase cells. In this sense, the increase in the system's temperature is only a statistical manifestation.

References

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